Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Choose Inputs and Outputs before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Choose Inputs and Outputs trace and record the first step you can explain without help.

Use W2: Run the **baseline** on three requests for **Requirement**s and a simple **baseline**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Choose Inputs and Outputs explanation against the supplied facts before discussing it.

Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build a baseline for a different catalogue

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Choose Inputs and Outputs target to a trace and a justified conclusion.

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Choose Inputs and Outputs case and show the deciding evidence.

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Choose Inputs and Outputs adds to Helpful Classroom Sorter and keep the

> evidence.

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Make a requirement checkable

> Make a requirement checkable. Apply the supplied Choose Inputs and Outputs facts and keep your reasoning.

Replace “suggest a good activity” with the time condition, quiet condition and **tie rule** used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Run the baseline on three requests

> Run the baseline on three requests. Apply the supplied Choose Inputs and Outputs facts and keep your reasoning.

Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Separate invalid input from no matching option

> Separate invalid input from no matching option. Apply the supplied Choose Inputs and Outputs facts and keep your reasoning.

Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the **baseline** record for each?

Explain why these are different outcomes even though neither produces an activity suggestion.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Build a baseline for a different catalogue

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a baseline for a different catalogue. Apply the supplied Choose Inputs and Outputs facts and keep your reasoning.

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Challenge an unfair comparison

> Challenge an unfair comparison. Apply the supplied Choose Inputs and Outputs facts and keep your reasoning.

Team A tests a prototype on five easy requests and the manual **baseline** on five different requests, then reports only the prototype's successful **output**s. Identify two changes needed for a fair comparison.

If both methods then meet all **requirement**s, what can and cannot be concluded about compatibility?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Choose Inputs and Outputs

> A fictional classroom chooses a storage bin for model parts. Explain the labelled step before checking the changed condition.

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

AI ACADEMY / AI-1/30

Choose Inputs and Outputs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate request

Count 6 is whole and within 1-12.



2. Check first bin

B1 capacity 5 is below 6; reject it.



3. Check second bin

B2 capacity 9 is at least 6; accept it.



4. Baseline output

Report B2 and stop; save the comparison evidence.

CHANGE ONE THING

Change request 6 to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace requests 10 and blank. Why should they not receive the same explanation?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional classroom chooses a storage bin for model parts.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report B2 and stop; save the comparison **evidence**.

1. **Baseline output**

2. Check first bin

3. Validate request

4. Check second bin

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Baseline output** Report B2 and stop; save the comparison **evidence**. The **baseline** is an ordered catalogue scan, not a preference ranking.

It makes expectations testable before a prototype is built. Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change request 6 to 5. Which conclusion is supported by the supplied facts?

1. The **baseline** returns B1.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **baseline** returns B1. Its

capacity equals the request, and at least includes equality.

Repair a missing building block

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

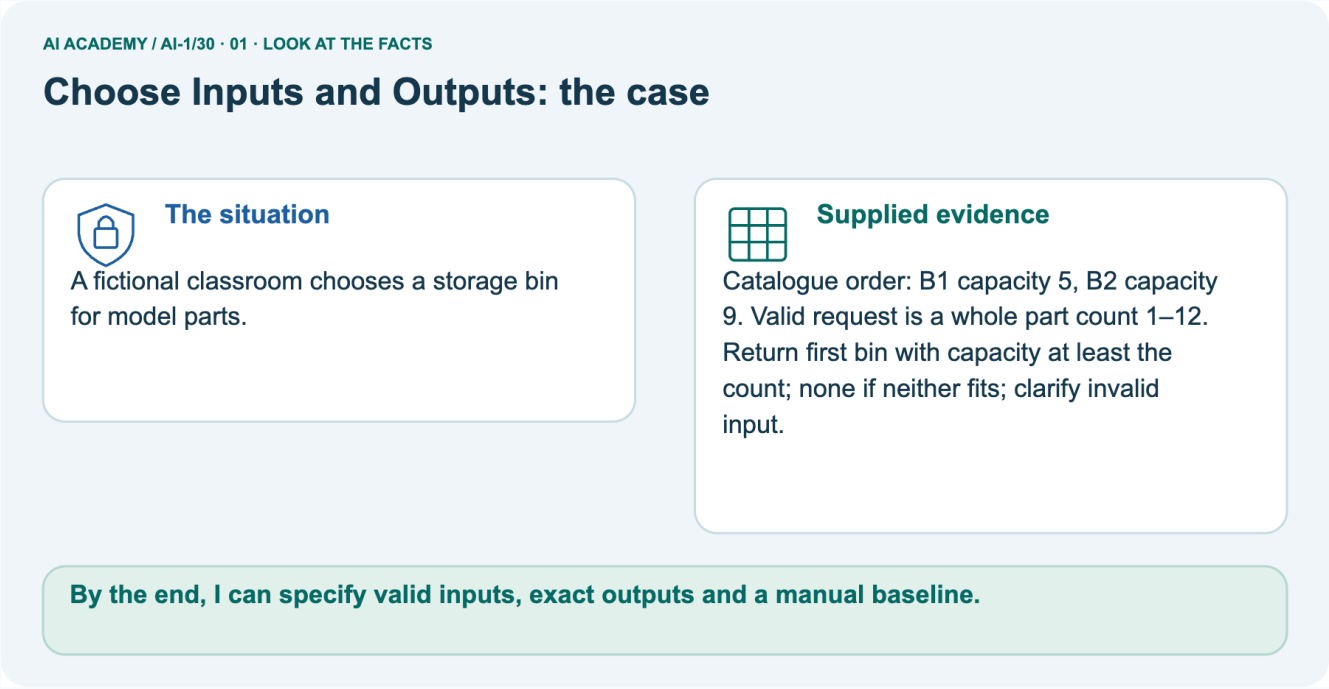
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Choose Inputs and Outputs picture from its labels, then record a relevant limit.

01. Choose Inputs and Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

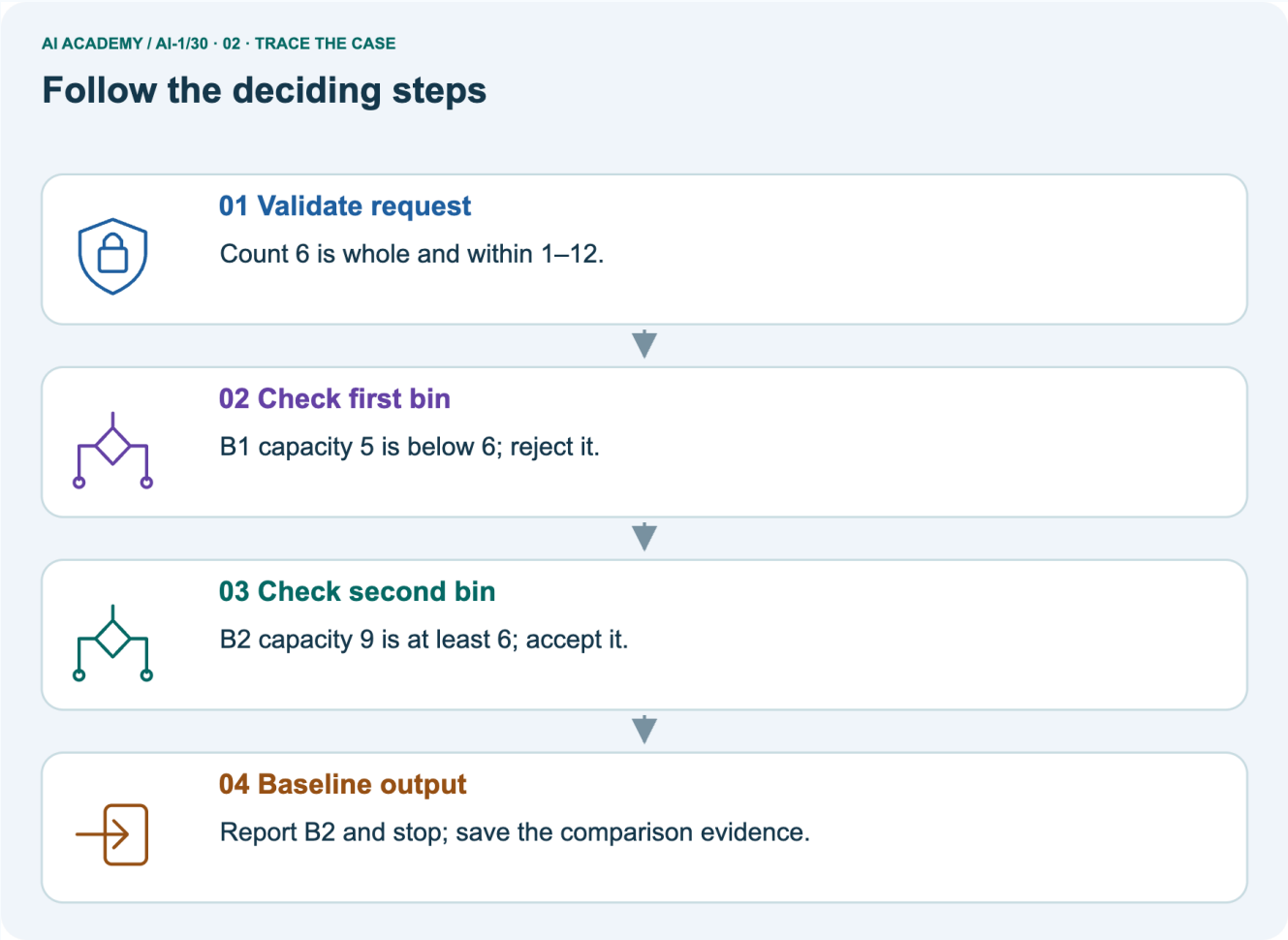
A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



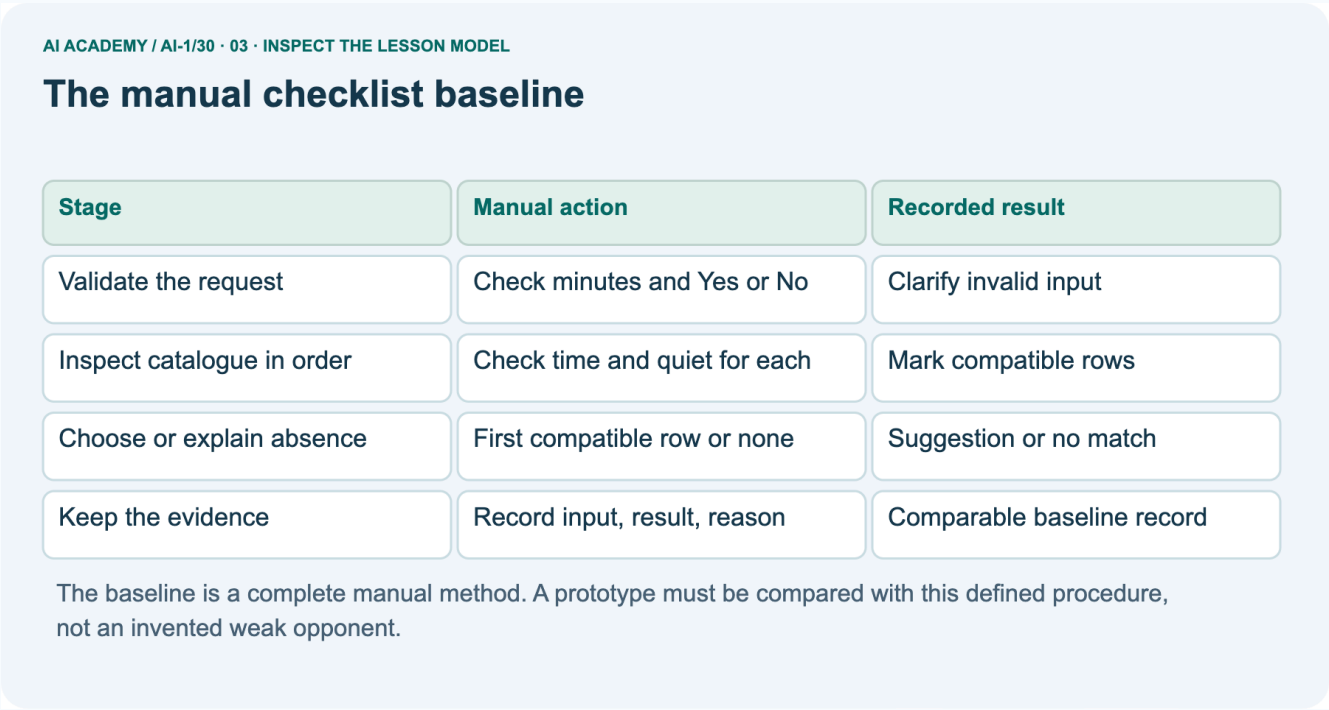
Follow the labelled picture with the matching explanation. Open the vector diagram

Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The manual checklist baseline



Follow the labelled picture with the matching explanation. Open the vector diagram

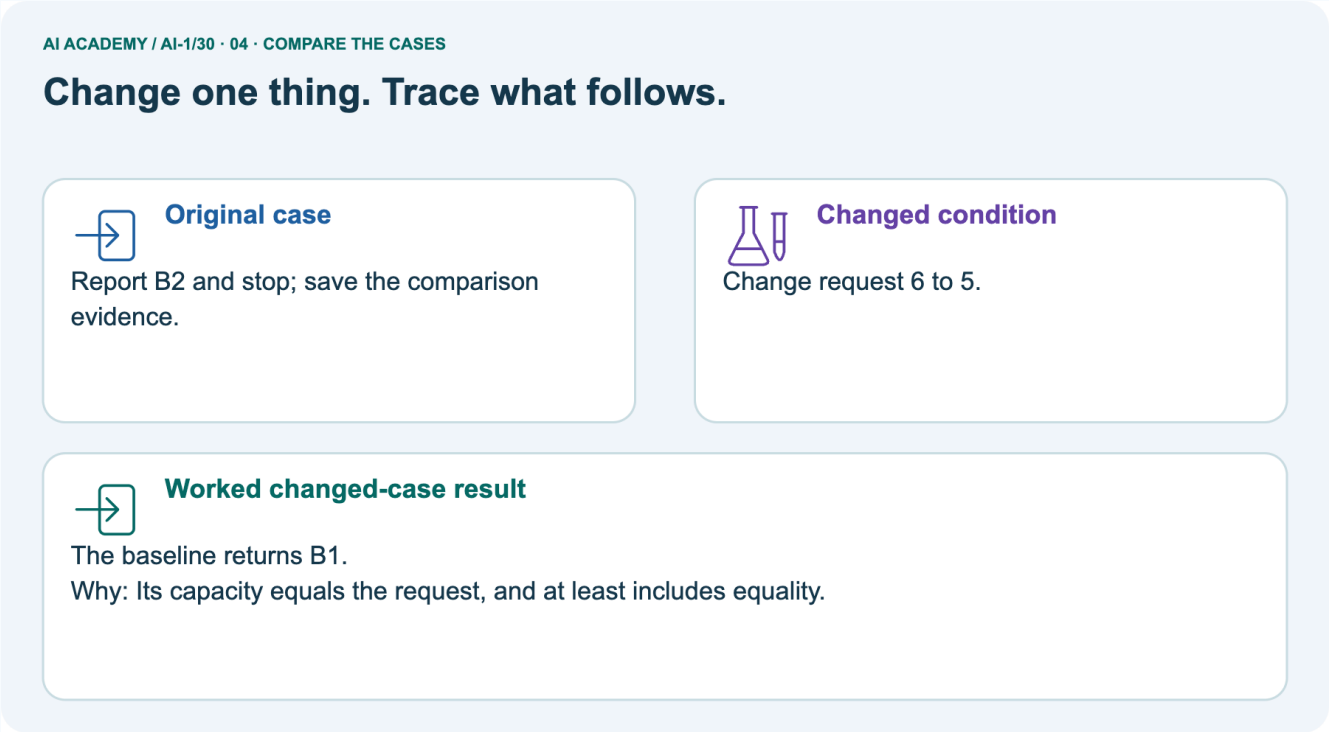
The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



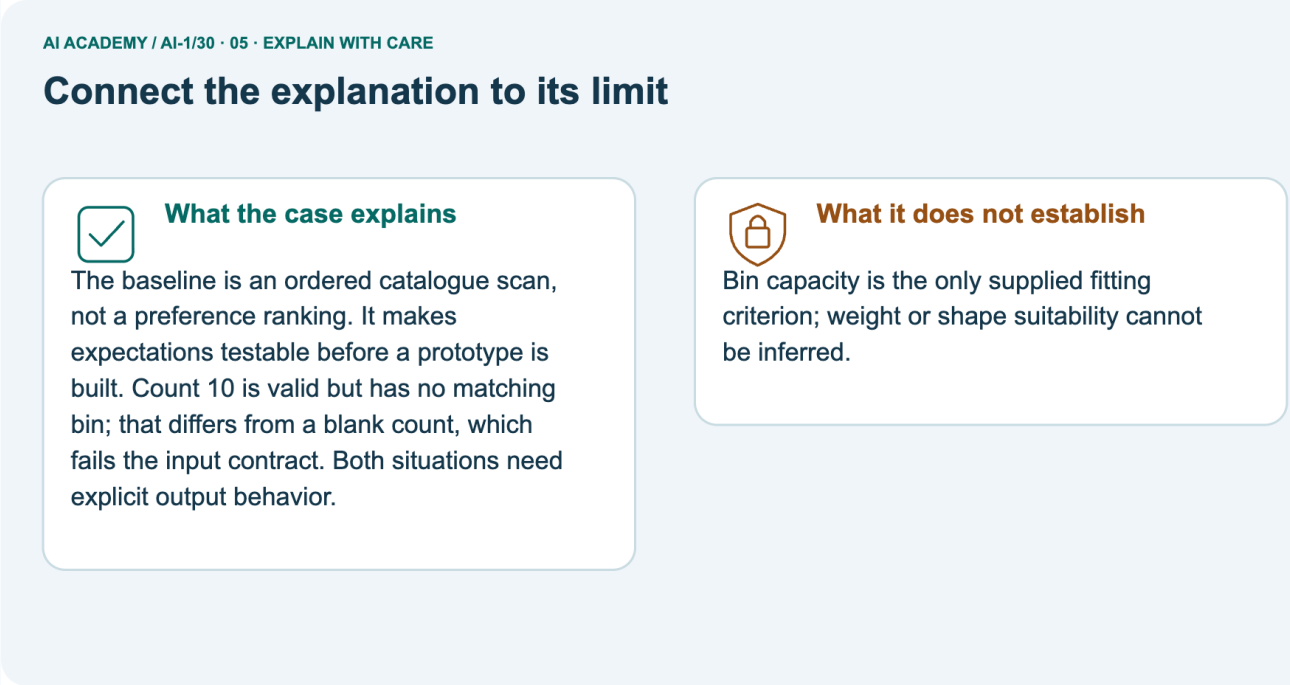
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

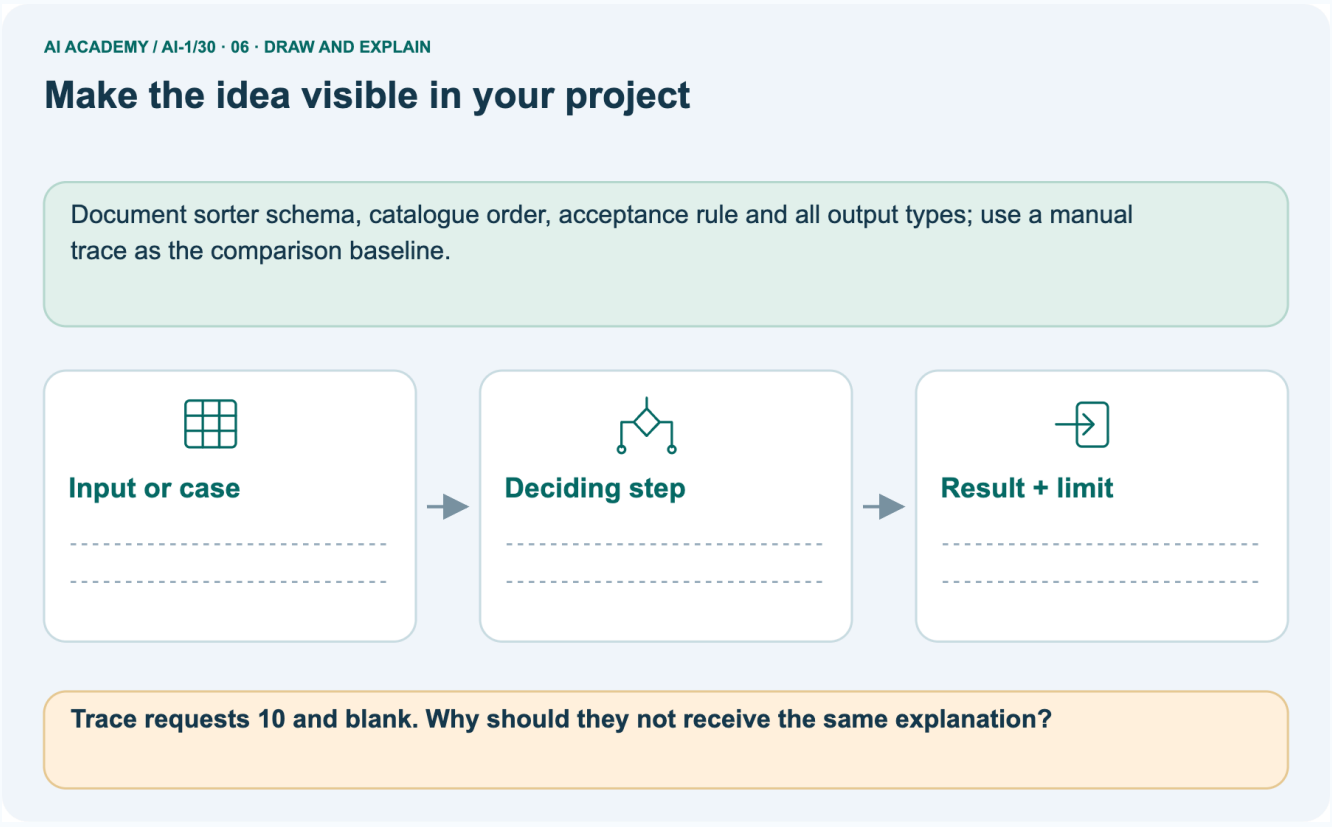
Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.